



Graphs in Machine Learning

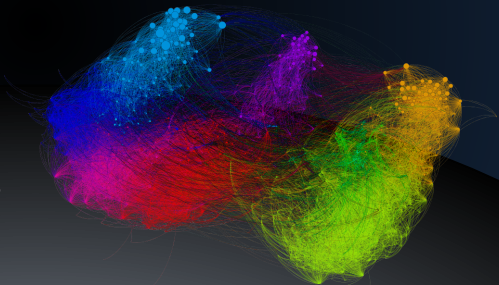
Online SSL Introduction

Problem Setup and Challenges

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Isara Labs

Partially based on material by: Branislav Kveton,
Mikhail Belkin, Jerry Zhu



Online SSL with Graphs

Offline learning setup

Given $\{\mathbf{x}_i\}_{i=1}^N$ from $\mathbb{R}^d$ and $\{y_i\}_{i=1}^{n_l}$, with $n_l \ll n$, find $\{y_i\}_{i=n_l+1}^N$ (transductive) or find f predicting y well beyond that (inductive).

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The cost and memory of the operations.

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What can we do?

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Proof?

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Why do we keep the multiplicities?

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- 5: **while** new unlabeled example $\mathbf{x}_t$ comes **do**
- 6: Add $\mathbf{x}_t$ to graph G
- 7: **if** # nodes $> k$ **then**
- 8: quantize G
- 9: **end if**

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- 9: **end if**
- 10: Update $\mathbf{L}_t$ of $G(\mathbf{V}\mathbf{W}\mathbf{V})$

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 - 10: Update $\mathbf{L}_t$ of $G(\mathbf{V}\mathbf{W}\mathbf{V})$
 - 11: Infer labels
 - 12: Predict $\hat{y}_t = \text{sgn}(\mathbf{f}_u(t))$
 - 13: **end while**
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<https://misovalko.github.io/mva-ml-graphs.html>